PART A – Dataset Description, Sample Design, and Summary Statistics

The data used in this analysis come from the Children's People and Nature Survey (C-PaNS), collected by Natural England between 2021 and 2023 across six waves of online data collection conducted during both school-term and school-holiday periods. The survey aims to monitor children's engagement with nature, wellbeing, and environmental attitudes. This report focuses on the association between children's connection to nature and their reported happiness, and 3 other explanatory variables; environmental attitudes, region of residence and period which were suggested to be related in the previously commissioned report.

Regarding the nature of the five variables, reported happiness (0–10) is treated as a continuous numeric variable because the scale is evenly spaced and supports meaningful calculations of means, standard deviations, and linear regression coefficients. Connection to nature (1–7) and environmental attitudes (1–5) are ordinal rating scales, but analysed as numeric variables since they approximate continuous distributions. Region is a nominal categorical variable with nine categories, and survey period (term-time vs school holiday) is a binary categorical variable. Categorical variables are summarised through proportions and entered into the regression model as indicator variables relative to a reference group. All variables are self-reported, which is typical for wellbeing research but introduces some measurement subjectivity.

According to sample design, quota-based sampling via Kantar's Profiles panel are used, with quotas aligned to population distributions for age, region, and ethnicity, including additional boosts for minority groups. The survey uses an online panel and therefore does not report a conventional response rate. The dataset applies design weights (adjusting for the number of eligible children in a household) and non-response weights (aligning the sample with population margins for age, gender, region, and ethnicity).

The original dataset contained 12,103 observations. A complete-case approach was applied, removing respondents with missing data on any analytic variable, resulting in a final sample of 11,796 children. Missingness checks suggested the data are Missing Completely At Random (MCAR), given low levels of item non-response. Although the dataset provides design and non-response weights to achieve population representativeness, these were not applied in the analysis because the task focuses on estimating associations rather than producing population estimates, and unweighted regression is appropriate under completecase MCAR assumptions.

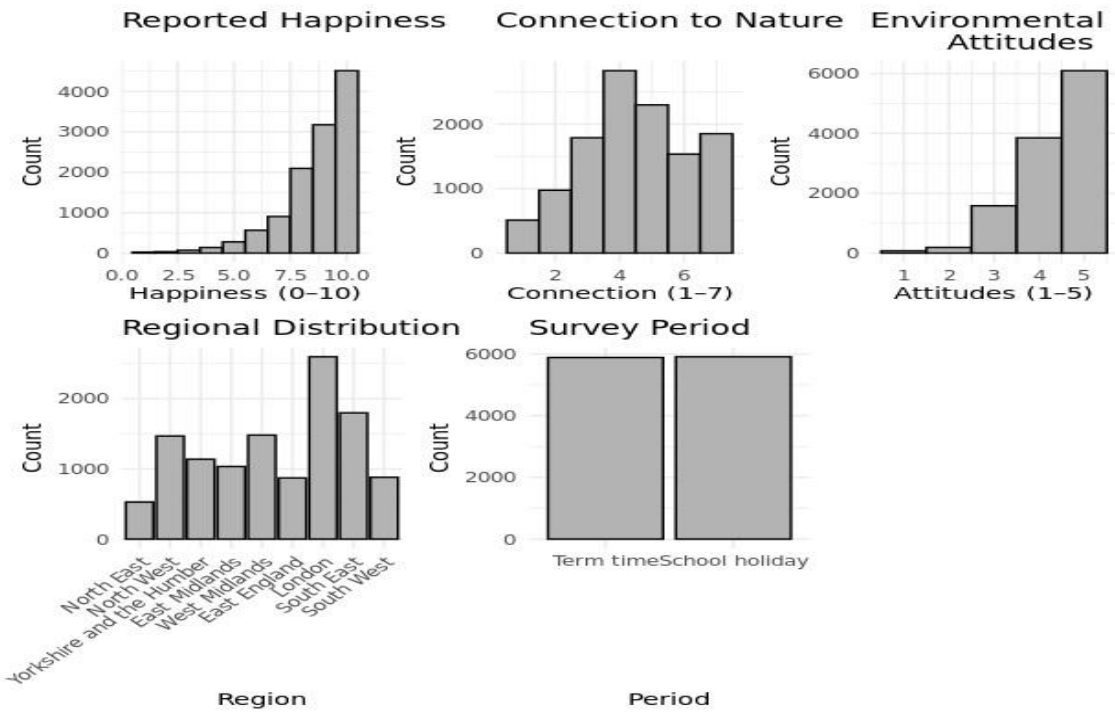
Summary statistics for the analytic sample are presented in Table 1 below. These show generally high levels of wellbeing (mean happiness = 8.68, SD = 1.54), moderate-to-high connection to nature (mean = 4.48, SD = 1.67), and environmental attitudes that are positively skewed towards stronger pro-environmental views (mean = 4.33, SD = 0.81). The distributions for region and survey period closely match the intended quota structure: regional representation remains balanced across England, with London, the South East, and the North West forming the largest subgroups, and data collection is evenly divided between term-time and school-holiday periods.

Table.1 Sample characteristics before and after complete-case analysis.

Sample size(N)= 12103 **Final sample size (N)= 11,796**			
Variables	**Original Proportion in % (counts)**	**Missing Proportion in % (counts)**	**Final Proportion in % (counts)**
Connection to nature 1 2 3 4 5 6 7	**Mean: 4.45** **SD: 1.67** 4.49 (543) 8.55 (1035) 15.24(1845) 24.15 (2923) 19.19 (2322) 12.81 (1550) 15.48 (1874)	0.09 (11)	**Final Mean: 4.48** **Final SD: 1.67** 4.33 (511) 8.27 (976) 15.17 (1790) 23.99 (2830) 19.50 (2300) 13.02 (1536) 15.71 (1853)
Reported happiness 1 2 3 4 5 6 7 8 9 10	**Mean: 8.67** **SD: 1.54** 0.21 (26) 0.28 (34) 0.60 (73) 1.25 (151) 2.35 (285) 4.88 (591) 7.68 (929) 17.75 (2148) 26.77 (3240) 38.02 (4602)	0.20(24)	**Final Mean: 8.68** **Final SD:1.54** 0.21 (25) 0.29 (34) 0.59 (70) 1.18 (139) 2.35 (277) 4.79 (565) 7.67 (905) 17.76 (2095) 26.92 (3175) 38.42 (4511)
Environmental attitudes 1 2 3 4 5	**Mean: 4.33** **SD: 0.81** 0.63 (76) 1.57 (190) 13.08 (1583) 31.88 (3859) 50.48 (6109)	2.36 (286)	**Final Mean: 4.33** **Final SD: 0.81** 0.64(76) 1.61 (190) 13.41 (1582) 32.66 (3852) 51.68 (6096)

Region North East North West Yorkshire and the Humber East Midlands West Midlands East England London South East South West	4.53 (548) 12.47 (1509) 9.66 (1169) 8.77 (1061) 12.59 (1524) 7.37 (892) 21.93 (2654) 15.15 (1834) 7.54 (912)	0.00 (0)	4.48 (528) 12.44 (1468) 9.65 (1138) 8.77 (1034) 12.56 (1481) 7.38 (871) 22.02 (2597) 15.24 (1798) 7.47 (881)
Period term time School holiday	49.97 (6048) 50.03 (6055)	0.00 (0)	49.9 (5886) 50.1 (5910)

Figure 1. Distributions of key variables in the analytic sample (n = 11,796).



To support the descriptive statistics, Figure 1 displays the distributions of all variables included in the analysis. The continuous variables, reported happiness, connection to nature, and environmental attitudes, show clear patterns: happiness scores are strongly right-skewed towards higher wellbeing, connection to nature shows a moderate right skew, and environmental attitudes are heavily clustered at the upper end of the scale. The categorical variables show expected sample patterns. Regional representation broadly reflects the intended quota structure, and the survey period is almost evenly split between term-time and

school-holiday responses. These distributions help contextualise the regression analysis by illustrating the variability and shape of the key analytic variables.

Regression analysis

Variables	Estimate	Std. Error	t-value	P-values
(intercept)	6.78	0.08	83.58	< 2e-16
Connection to Nature	0.13	0.01	14.52	< 2e-16
Environmental attitudes	0.34	0.02	18.92	< 2e-16
Region(reference= London)				
North East	-0.03	0.07	-0.46	0.64
North West	-0.09	0.05	-1.85	0.06
Yorkshire and the Humber	-0.07	0.05	-1.40	0.16
East Midlands	-0.05	0.06	-0.97	0.33 0.11
West Midlands	-0.08	0.05	-1.61	0.02
East England	-0.13	0.06	-2.24	0.001
South East	-0.15	0.05	-3.22	0.15
South West	-0.08	0.06	-1.44	
			- 4.15	
Period(reference= term-time**)**				
School holiday	-0.11	0.03	-4.15	3.35e-05
Residual standard error: 1.481 **Adjusted R-squared**: 0.07035 **F-statistic**: 82.15 (p-value: < 2.2e-16)				

Table 2. Multiple linear regression results predicting children's reported happiness (N=11,796). **Significance levels: ***$p < 0.05$ (), **$p < 0.01$ (), *$p < 0.001$ ()**

Checks

Diagnostic checks indicated no influential outliers (all Cook's distances far below 0.5). VIF and GVIF values were all close to 1, showing no evidence of multicollinearity. Since multicollinearity checks indicated that predictors were largely independent, the estimated effects can be interpreted as distinct contributions.

Regression Interpretation

A multiple linear regression model was estimated to assess whether children's happiness (0–10 scale) is associated with their connection to nature, controlling for environmental attitudes, region, and the survey period. All effects are interpreted relative to the reference categories: London (region) and term-time (period).

Connection to nature

Holding all other variables constant, each one-point increase in connection to nature is associated with a 0.13-point increase in reported happiness ($t = 14.52$, $p < 0.001$). This indicates a small but statistically significant positive association.

Environmental attitudes

Environmental attitudes show an even stronger association with happiness: a one-unit increase corresponds to a 0.34-point increase in happiness ($t = 18.92$, $p < 0.001$), adjusting for other variables. This suggests that pro-environmental attitudes are meaningfully linked to children's wellbeing.

Region (reference: London)

Two regions show statistically significant differences relative to London. Children in the East of England report happiness scores 0.13 points lower ($p = 0.025$), while those in the South East report scores 0.15 points lower ($p = 0.001$), holding other factors constant. Coefficients for all other regions are also negative but not statistically significant. Overall, the model suggests modest regional variation in wellbeing once demographic and attitudinal factors are controlled for.

Survey period (reference: term-time)

Children surveyed during school holidays report 0.11-point lower happiness relative to those surveyed in term-time ($t = –4.15$, $p < 0.001$). Although small, this indicates that the timing of data collection may influence self-reported wellbeing.

Overall model fit

The model explains a modest proportion of variation in happiness (adjusted $R^2 = 0.070$). This is expected given the many unobserved influences on wellbeing. However, the overall F-test ($F = 82.15$, $p < 0.001$) indicates that the predictors jointly explain significant variation in children's happiness.

Conclusion

Overall, the results show that children who feel more connected to nature and who hold stronger pro-environmental attitudes report slightly higher levels of happiness, independent of demographic and contextual factors. Small but statistically significant differences are also observed across some regions and between term-time and holiday data collection, although these effects are modest once controls are included.

A key limitation is the use of complete-case analysis, which assumes that missing responses are Missing Completely At Random (MCAR). If this assumption is violated, estimates may be biased. In practice, methods such as multiple imputation could be used to assess and reduce potential bias.

PART B: Regression Interpretation

Model 2 includes an interaction between neighbourhood co-ethnic density in 2001 and individual ethnic minority groups. The model estimates whether the association between coethnic density and mental health differs for each ethnic minority group compared with the White British reference group. GHQ-12 scores are interpreted as higher values indicating worse mental health, meaning negative coefficients reflect *better* mental health.

Interpretation of the Interaction Terms (β_3)

For the i) White Other group, the interaction coefficient of −0.187 (95% CI [−0.288, −0.085]) suggests that as co-ethnic density increases, they experience greater improvement in mental health relative to the White British group. This is statistically significant because the confidence interval does not include zero.

ii) Mixed group, the coefficient of −0.082 (95% CI [−0.211, 0.047]) indicates a similar, but weaker, pattern; however, the result is not statistically significant.
The Indian group also shows a significant beneficial difference (−0.134, 95% CI [−0.228, −0.039]), suggesting that higher co-ethnic density is linked to relatively better mental health for Indian individuals compared with White British.

iii) Pakistani group (−0.069, 95% CI [−0.211, 0.047]) and Caribbean group (−0.154, 95% CI [−0.377, 0.070]) both show negative but non-significant coefficients, implying similar trends but insufficient evidence to confirm a significant difference.

In contrast, the Bangladeshi group (0.117, 95% CI [−0.038, 0.272]) shows a positive but nonsignificant coefficient, suggesting a slightly weaker or reversed association between co-ethnic density and mental health compared with White British individuals as although their mental health worsens, it doesn't seem to be a statistically significant change.

Finally, both the African (−0.132, 95% CI [−0.292, 0.028]) and Other (−0.051, 95% CI [−0.172, 0.069]) groups display small, non-significant negative coefficients, suggesting no strong evidence that co-ethnic density affects their mental health differently from the White British group.

Conclusion;

Overall, Model 2 shows that the relationship between co-ethnic density and mental health varies only modestly across ethnic groups. Although most interaction terms are negative, suggesting slightly better mental health with higher co-ethnic density, only the White Other and Indian groups show statistically significant differences from the White British reference

group. For all other minority groups, confidence intervals include zero, indicating no clear evidence of a distinct effect. Thus, the results provide limited support for stronger protective effects of co-ethnic density among specific groups, while suggesting that the association is broadly similar for most ethnic populations.

Regarding data approach limitations, co-ethnic density is based on 2001 Census data rather than on conditions at the time the mental health data were collected. This assumes that neighbourhood ethnic composition stayed the same over time, which may not be true and could reduce the accuracy of the estimated effects.

Acknowledgements;

I acknowledge the use of ChatGPT, to improve my code and my writing.

R- Appendix

```r
#POLS0010 ASSIGNMENT1 R SCRIPT

#RESET ENVIRONMENT rm(list = ls()) \# Clear environment graphics.off()
# Close any open plots cat("\\014") \# Clear console

# STEP 1: LOAD PACKAGES

library(tidyverse)
library(ggplot2)
library(patchwork)

# STEP 2: IMPORT DATA

df <-
  read_csv("c-pans_waves_1-6_holiday_2021_term_2023_open_access.csv")
```

```r
# STEP 3: SELECT VARIABLES

# Y = CS_Wellbeing_happy (happiness, 0--10)

# X = CS_Q11 (connection to nature, 1--7)

# x2 = CS_Q9 (environmental attitudes, 1--5)

# x3 = CS_S_Region (region)

# x4 = Period (term time / school holiday)

df \<- df %\>% select( CS_Wellbeing_happy, \# Y CS_Q11, \# X CS_Q9, \#
                       control CS_S_Region, \# control Period \# control )

                       # STEP 4: RECODING VARIABLES

                       ## 4.1 Recode wellbeing (fix any 11s to 10, keep 0--10 scale)

                       df \<- df %\>% mutate( CS_Wellbeing_happy = ifelse(CS_Wellbeing_happy \>
                                                                          10, 10, CS_Wellbeing_happy) )

                       ## 4.2 Recode Period (1 = term time, 2 = school holiday)

                       df \<- df %\>% mutate( Period = factor( Period, levels = c(1, 2), labels
                                                   = c("Term time", "School holiday") ) )

                       ## 4.3 Recode Region (1--9 → named UK regions)

                       df \<- df %\>% mutate( CS_S_Region = factor( CS_S_Region, levels = 1:9,
                                                                 labels = c( "North East", "North West", "York
shire and the Humber",
                                                                             "East Midlands", "West Midlands",
"East England", "London", "South
East", "South West" ) ) )
```

```
# check of distinct values in each variable

df %\>% summarise(across(everything(), n_distinct))

#RECODING CHECK

# 1. Check wellbeing (should be 0--10 only)

table(df\$CS_Wellbeing_happy, useNA = "ifany")

# 2. Check Period (should show "Term time" and "School holiday")

table(df\$Period, useNA = "ifany")

# 3. Check Region (should show 9 named regions)

table(df\$CS_S_Region, useNA = "ifany")

# 4. Check CS_Q11 (connection to nature) 1--7?

table(df\$CS_Q11, useNA = "ifany")

# 5. Check CS_Q9 (thoughts on importance of environment) 1--5?

table(df\$CS_Q9, useNA = "ifany")

# STEP 5: MISSINGNESS CHECK

# Create a list of variables you want to check

vars \<- c("CS_Wellbeing_happy", "CS_Q11", "CS_Q9", "CS_S_Region",
           "Period")

# Total sample size

total_n \<- nrow(df)
```

```
# Build missingness summary

missing_summary \<- df %\>% summarise(across(all_of(vars),
                                             \~sum(is.na(.)), .names = "Missing\_{col}"))
%\>%
  pivot_longer(everything(), names_to = "Variable", values_to =
               "Missing_N") %\>% mutate( Variable = gsub("Missing\_", "", Variable),
                                         Total_N = total_n, Missing_Pct = round((Missing
_N / total_n) \* 100, 2),
                                         Valid_N = Total_N - Missing_N, Valid_Pct = roun
d((Valid_N / Total_N) \*
100, 2) )

missing_summary

#STEP 6: Creating complete-case dataset

df_complete \<- df %\>% drop_na(CS_Wellbeing_happy, CS_Q11, CS_Q9,
                                CS_S_Region, Period)

final_N \<- nrow(df_complete) final_N \# should return 11796

# STEP 7: DESCRIPTIVE STATISTICS

# 1) WELLBEING- Y

# Wellbeing descriptive statistics

wellbeing_stats \<- df %\>% summarise( n =
                                         sum(!is.na(CS_Wellbeing_happy)), mean = mean(CS_W
ellbeing_happy, na.rm =
                                                                                          TR
UE), sd = sd(CS_Wellbeing_happy, na.rm = TRUE), median =
                                         median(CS_Wellbeing_happy, na.rm = TRUE), min = m
in(CS_Wellbeing_happy,
na.rm = TRUE), max = max(CS_Wellbeing_happy, na.rm = TRUE) )

wellbeing_freq \<- df %\>% count(CS_Wellbeing_happy) %\>% mutate( pct =
                                                                  round(n / sum(n) \* 10
0, 2) )

wellbeing_stats wellbeing_freq
```

```
#2) CONNECTION TO NATURE(X) \# Connection to Nature (CS_Q11) conn_stats
\<- df %\>% summarise( n = sum(!is.na(CS_Q11)), mean = mean(CS_Q11,
                                                                    na.rm = TRUE), sd = sd(CS_Q11,
na.rm = TRUE), median = median(CS_Q11,

na.rm = TRUE), min = min(CS_Q11, na.rm = TRUE), max = max(CS_Q11, na.rm

= TRUE) )

conn_freq \<- df %\>% count(CS_Q11) %\>% mutate( pct = round(n / sum(n)
                                                                    \* 100, 2) )

conn_stats conn_freq

#DECRIPTIVE STATISTICS FOR FINAL COUNTS final_stats \<- df_complete %\>%
summarise( final_mean_happy = mean(CS_Wellbeing_happy), final_sd_happy =
             sd(CS_Wellbeing_happy),

          ```
          final_mean_conn  = mean(CS_Q11),
          final_sd_conn    = sd(CS_Q11),

          final_mean_env   = mean(CS_Q9),
          final_sd_env     = sd(CS_Q9)
          ```

)

final_stats
```

```
#VISUALISATIONS- Descriptive plots p1 \<- ggplot(df_complete, aes(x =
CS_Wellbeing_happy)) + geom_histogram(binwidth = 1, fill = "grey70",
                                                    color = "black") + labs(title = "Reported Happines
s", x = "Happiness
(0--10)", y = "Count") + theme_minimal()

p2 \<- ggplot(df_complete, aes(x = CS_Q11)) + geom_histogram(binwidth =
                                                    1, fill = "grey70", color = "black") + labs(title
= "Connection to
Nature", x = "Connection (1--7)", y = "Count") + theme_minimal()

p3 \<- ggplot(df_complete, aes(x = CS_Q9)) + geom_histogram(binwidth =
                                                    1, fill = "grey70", color = "black") + labs(title =
"Environmental
Attitudes", x = "Attitudes (1--5)", y = "Count") + theme_minimal()

p4 \<- ggplot(df_complete, aes(x = CS_S_Region)) + geom_bar(fill =
                                                    "grey70", color = "black") + labs(title = "Regional
Distribution", x =
                                                                                    "Region", y =
"Count") + theme_minimal() + theme(axis.text.x =

element_text(angle = 45, hjust = 1))

p5 \<- ggplot(df_complete, aes(x = Period)) + geom_bar(fill = "grey70",
                                                    color = "black") + labs(title = "Survey Period", x = "Peri
od", y =
                                                                    "Count") + theme_minimal()

# Top row: 3 plots

top_row \<- p1 \| p2 \| p3

# Bottom row: 2 plots (with empty space for balance)

bottom_row \<- p4 \| p5 \| plot_spacer()

# Combine rows

(top_row) / (bottom_row)

#MULTICOLLINEARITY CHECK #install VIF package library(car)

#fit a temporary model for complete case model_temp \<- lm(
CS_Wellbeing_happy \~ CS_Q11 + CS_Q9 + CS_S_Region + Period, data =
  df_complete )

vif(model_temp)
```

```r
#CHECK FOR LINEARITY between continuous predictors and outcome
library(ggplot2)

# Happiness vs Connection to Nature

ggplot(df_complete, aes(x = CS_Q11, y = CS_Wellbeing_happy)) +
  geom_point(alpha = 0.2) + geom_smooth(method = "lm") + labs(title =
                                                                "Linearity Check: Happiness vs Connection to Natu
re")

# Happiness vs Environmental Attitudes

ggplot(df_complete, aes(x = CS_Q9, y = CS_Wellbeing_happy)) +
  geom_point(alpha = 0.2) + geom_smooth(method = "lm") + labs(title =
                                                                "Linearity Check: Happiness vs Environmental Atti
tudes")

#SET REFERENCE GROUPS AS LONDON, AND TERM TIME
df_complete$CS_S_Region <- relevel(df_complete$CS_S_Region, ref =
                                     "London") df_complete$Period <- relevel(df_complete$Period, ref = "Term
time")

## FINAL REGRESSION

model_final \<- lm(CS_Wellbeing_happy \~ CS_Q11 + CS_Q9 + CS_S_Region +
                     Period, data = df_complete) summary(model_final)

#Check for outliers model_temp \<- lm(CS_Wellbeing_happy \~ CS_Q11 +
CS_Q9 + CS_S_Region + Period, data = df_complete) plot(model_temp, which
                                                        = 4) \# Cook's distance
```